



INDIAN STATISTICAL INSTITUTE

Bachelor of Statistical Data Science

Intermediate Microeconomics

Notes based on Hal R. Varian, Intermediate Microeconomics: A Modern Approach

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Contents

1	Budget Constraint	2
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Budget Constraint

Let $X = (x_1, x_2)^t$ denote the **consumption bundle** of the consumer such that they chooses to consume x_1 amount of good 1 and x_2 amount of good 2

Let $P = (p_1, p_2)^t$ be the observed prices of those two goods respectively and let m be the amount of money the consumer has to spend

THE BUDGET CONSTRAINT

Thus, we have the following budget constraint of the consumer:

$$p_1x_1 + p_2x_2 \leq m \iff X^tP \leq m$$

■ BUDGET SET

The set of *affordable consumption* at price P

■ TWO GOODS ASSUMPTION

We consider x_1 as the amount of good that is of the interest of study and x_2 as everything else that the consumer can consume. It is convenient to take x_2 as the amount of dollars the consumer has to spend on everything else. Thus, $p_2 = 1$ since the cost of 1 dollar is 1 dollar itself. Thus, we arrive at the following budget constraint

$$p_1x_1 + x_2 \leq m$$